



OpenCore

Reference Manual (1.0.~~7~~.8)

[2026.09.13]

- **Windows** — Windows (128x128).
- [OpticalDrive — Optical disc drive \(128x128\)](#).
- **Other** — Custom entry (see **Entries**, 128x128).
- **ResetNVRAM** — Reset NVRAM system action or tool (128x128).
- **Shell** — Entry with UEFI Shell name for e.g. **OpenShell** (128x128).
- **Tool** — Any other tool (128x128).

Predefined labels are saved in the `\EFI\OC\Resources\Label` directory. Each label has `.1b1` or `.12x` suffix to represent the scaling level. Full list of labels is provided below. All labels are mandatory.

- **EFIBoot** — Generic OS.
- **Apple** — Apple OS.
- **AppleRecv** — Apple Recovery OS.
- **AppleTM** — Apple Time Machine.
- **Windows** — Windows.
- **Other** — Custom entry (see **Entries**).
- **ResetNVRAM** — Reset NVRAM system action or tool.
- **SIPDisabled** — Toggle SIP tool with SIP disabled.
- **SIPEnabled** — Toggle SIP tool with SIP enabled.
- **Shell** — Entry with UEFI Shell name (e.g. **OpenShell**).
- **FirmwareSettings** — Firmware settings menu entry.
- **Tool** — Any other tool.

Note: All labels must have a height of exactly 12 px. There is no limit for their width.

Label and icon generation can be performed with bundled utilities: `disklabel` and `icnspack`. Font is Helvetica 12 pt times scale factor.

Font format corresponds to AngelCode binary BMF. While there are many utilities to generate font files, currently it is recommended to use `dpFontBaker` to generate bitmap font (using `CoreText` produces best results) and `fonverter` to export it to binary format.

11.5 OpenRuntime

OpenRuntime is an OpenCore plugin implementing `OC_FIRMWARE_RUNTIME` protocol. This protocol implements multiple features required for OpenCore that are otherwise not possible to implement in OpenCore itself as they are needed to work in runtime, i.e. during operating system functioning. Feature highlights:

- NVRAM namespaces, allowing to isolate operating systems from accessing select variables (e.g. **RequestBootVarRouting** or **ProtectSecureBoot**).
- Read-only and write-only NVRAM variables, enhancing the security of OpenCore, Lilu, and Lilu plugins, such as **VirtualSMC**, which implements **AuthRestart** support.
- NVRAM isolation, allowing to protect all variables from being written from an untrusted operating system (e.g. **DisableVariableWrite**).
- UEFI Runtime Services memory protection management to workaround read-only mapping (e.g. **EnableWriteUnprotector**).

11.6 OpenLegacyBoot

OpenLegacyBoot is an OpenCore plugin implementing `OC_BOOT_ENTRY_PROTOCOL`. It aims to detect and boot legacy installed operating systems on supported systems, such as OpenDuet and Mac models capable of legacy booting.

Usage:

- Add **OpenLegacyBoot.efi** and also optionally (see below) **OpenNtfsDxe.efi** to the `config.plist Drivers` section.
- Install Windows or another legacy operating system as normal if this has not been done earlier – **OpenLegacyBoot** is not involved in this stage and may be unable to boot from installation media such as a USB device.
- Reboot into OpenCore: the installed legacy operating system should appear and boot directly from OpenCore when selected.

AppleHDA kext will function correctly only if TCSEL register is configured to use TC0 traffic class. Refer to Intel I/O Controller Hub 9 (ICH9) Family Datasheet (or any other ICH datasheet) for more details about this register.

Note: This option is independent from AudioSupport. If AppleALC is used it is preferred to use AppleALC alctcsel property instead.

11. SetupDelay

Type: plist integer

Failsafe: 0

Description: Audio codec reconfiguration delay in milliseconds.

Some codecs require a vendor-specific delay after the reconfiguration (e.g. volume setting). This option makes it configurable. A typical delay can be up to 0.5 seconds.

11.16 Drivers Properties

1. Arguments

Type: plist string

Failsafe: Empty

Description: Some OpenCore plugins accept optional additional arguments which may be specified as a string here.

2. Comment

Type: plist string

Failsafe: Empty

Description: Arbitrary ASCII string used to provide human readable reference for the entry. Whether this value is used is implementation defined.

3. Enabled

Type: plist boolean

Failsafe: false

Description: If false this driver entry will be ignored.

4. HideVerbose

Type: plist boolean

Failsafe: false

Description: If true text output for this driver will be disabled.

5. LoadEarly

Type: plist boolean

Failsafe: false

Description: Load the driver early in the OpenCore boot process, before NVRAM setup.

Note: Do not enable this option unless specifically recommended to do so for a given driver and purpose.

6. Path

Type: plist string

Failsafe: Empty

Description: Path of file to be loaded as a UEFI driver from OC/Drivers directory.

11.17 Input Properties

1. KeyFiltering

Type: plist boolean

Failsafe: false

Description: Enable keyboard input sanity checking.

Apparently some boards such as the GA Z77P-D3 may return uninitialised data in EFI_INPUT_KEY with all input protocols. This option discards keys that are neither ASCII, nor are defined in the UEFI specification (see tables 107 and 108 in version 2.8).

2. KeyForgetThreshold

Type: plist integer